Yasemin Ozkut

Master of Science | Computer Vision Engineer Intern | AI/ML & Embodied AI New Grad (May 2026)
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Education

MS **The Ohio State University**, Electrical and Computer Engineering — Aug 2024 – May 2026
Focus: Computer Vision, Machine Learning, Deep Learning — Columbus, OH, USA

BS **Sabanci University**, Computer Science and Engineering — Sept 2019 – Feb 2024
Istanbul, Turkey

Skills

Languages: Python (4+ years hands-on, 2+ year professional), C++, SQL, JavaScript

Frameworks & Libraries: PyTorch, PyTorch Lightning, TensorRT, TensorBoard, ONNX, YOLO, OpenCV, Unsloth, Hugging Face (Accelerate, Datasets, Transformers), MONAI, LangGraph, Pandas, NumPy, Scikit-learn, React, Hydra, Flask

Tools: Git, GitHub Actions (CI/CD), CUDA, SLURM, Linux, Claude Code, Cursor, Reachy Mini SDK, Docker, PostgreSQL

Experience

Ubihere, Computer Vision Engineer Intern – Columbus, OH — Aug 2025 – Present

- Wrapped research-stage person re-identification modules into a GPU-accelerated Flask microservice with REST API, containerized with Docker and NVIDIA Container Toolkit; now integrated into core product.
- Engineered GPU batching pipeline and parallelized image I/O, reducing inference latency by 67%.
- Architected a widget-based dashboard in React, replacing hardcoded per-client builds with a single shared codebase and configurable widget library, adopted in the company's primary client-facing product featured in sales demos.
- Trained YOLOv8 on a multi-GPU SLURM setup for multi-class traffic vehicle detection, achieving 84% mAP50, upgrading to YOLO26 for improved accuracy and edge inference efficiency across 60+ city traffic cameras.

PCVLab (Computer Vision Lab), Graduate Research Assistant, ML/CV – Columbus, OH — Aug 2024 – Aug 2025

- Benchmarked 8 CNN/Transformer architectures for computer vision-based ocular ultrasound video classification across 5 binary tasks, training 40+ models on multi-GPU via SLURM using Lightning-Hydra and MONAI.
- Designed ratio-based top-k temporal pooling with L2 norm frame importance scoring, dynamically adapting to variable temporal dimensions across all 8 architectures, eliminating manual k configuration per architecture.
- Developed two-stage diagnostic pipeline for retinal detachment (97.4% accuracy) and macular status (88.2% accuracy) classification; published ERDES dataset and paper (10k+ Hugging Face downloads).
- Fine-tuned multi-modal VLM LLaMA-3.2-11B-Vision via 4-bit QLoRA (Unsloth/TRL) on 100K+ MIMIC-CXR chest X-rays for radiology report generation, on HPC (SLURM), reducing loss by 79%; BLEU-1: 28.1, ROUGE-1: 39.5.

DAI-Labor, AI/ML Researcher Intern – Berlin, Germany — July 2023 – Sept 2023

- Developed an end-to-end real-time detection pipeline by annotating 1,235 custom images to extend a German grocery dataset; trained a YOLOv8 model achieving 87.2% mAP50 performance.
- Integrated Keras-OCR to extract German-language labels and benchmarked performance against YOLOv8 detection.

Projects

Budi Desktop Robot Companion, Embodied AI, Team Project — Feb 2026

- **Top 3 Winner ($1,000), Embodied AI Hackathon 2026** – by Seeed Studio, NVIDIA, Hugging Face, and Pollen Robotics.
- Developed a desktop productivity companion on Reachy Mini Lite Robot + Jetson Orin Nano running Cosmos Reason2 VLM for conversations, Faster Whisper for STT, and Microsoft Edge TTS; all on-device except TTS.
- Implemented reminder system with voice-triggered alerts and focus mode for distraction blocking, Jetson Orin Nano integration debugging, and VLM Docker hosting setup.

Judgy Reachy, Embodied AI, Personal Project | Hugging Face — Jan 2026

- **5th place of 75, NVIDIA GTC 2026 Golden Ticket Contest** – Hugging Face x Pollen Robotics Reachy Mini Robot track.
- Built a real-time phone detection pipeline that delivers funny robot feedback using YOLO26m with TensorRT (2.69x speedup), Llama 3.1-8B (Groq API) with 9 personality profiles, ElevenLabs TTS, and expressive robot motion.
- Published as an open-source Hugging Face Space with CI/CD via GitHub Actions, supporting MuJoCo simulation,

physical robot, and standalone WebGPU browser demo (ONNX Runtime + Transformers.js).

Pollen Robotics Open Source Contributions, Reachy Mini SDK (1k+ stars), Reachy Mini Conversation App (160+ stars) — Jan 2026

- Fixed community app install bug in Reachy Mini SDK by adding metadata persistence and Python package entry point matching (PR merged in v1.2.13).
- Fixed missing auto-enable of Gradio UI flag in Reachy Mini Conversation App, enabling simulation mode without a physical robot (PR merged); contributing dynamic dance description loading enhancement (open PR).

Agentic Person Re-ID via LangGraph and VLM Reasoning, Personal Project — July 2025

- Built a LangGraph pipeline integrating YOLOv8, Qwen2.5-VL-3B, and Qwen2.5-7B for person detection, description generation, and identity assignment across frames/videos.
- Designed structured JSON prompts to extract rich semantic attributes (hair, clothing, accessories, face shape) and implemented persistent memory for global ID tracking.
- Enabled explainable matching with logged reasoning, confidence scores, and memory updates for re-identification.

Serious Game for Children with Cerebral Palsy , Sabanci University — Feb 2023 – Feb 2024

- Developed a serious game in Unity using C# for balance physiotherapy sessions in children with Cerebral Palsy.
- Collected live data from Wii Balance Board for each game session and stored it into MongoDB.
- Generated visual reports from the processed data using Python and Flask.

Spotify Music Artist Success Collaboration Network, Sabanci University — Feb 2023 – June 2023

- Collected and preprocessed Spotify and Kaggle data to build a collaboration network consists of 8.6K artists and 13.3K connections using Python/NetworkX, incorporating genre, popularity, and centrality metrics.
- Applied clustering and centrality algorithms to identify key influencers, and visualized the network with Gephi for success analysis.

Publications

ERDES: A Benchmark Video Dataset for Retinal Detachment and Macular Status Classification in Ocular Ultrasound, *Nature Scientific Data* Website | Hugging Face | ArXiv — In revision July 2025
Yasemin Ozkut, Pouyan Navard, Srikar Adhikari, Elaine Situ-LaCasse, Josie Acuña, Adrienne A Yarnish, Alper Yilmaz

Activities & Interests

Hiphop Dancer (2015–Present); **Board Member**, SuDance Club, Sabancı University (2019–2023); **Contestant**, Peak Games Unithon (2022); **Tutor**, Civic Involvement Project – 4th-grade students (2020); **Winter Sports Enthusiast** (Skiing, Ice Skating, Snowboarding)